Why Proof-of-Authority Consensus Is Ill-Suited for Commercial-Bank-Operated DLT-Based Financial-Market Infrastructures

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Distributed ledger technology (DLT) has been widely proposed as a foundation for interbank settlement, cross-border payments, and tokenized asset markets in the future financial-market infrastructures. In this context, systems are typically permissioned and rely on Proof-of-Authority (PoA) or similar consensus mechanisms to achieve scalability and operational efficiency.

This paper identifies structural governance tensions in PoA-based, commercial-bank operated DLT systems by analyzing the role of PoA through the lens of institutional market structure and economic incentives for participating or operating commercial banks. We argue that the institutional viability of authority-based consensus depends less on technical performance than on the identity, mandate, and incentives of the authority controlling transaction ordering, inclusion, and technical finality. PoA with a single validating entity introduces strategic dependence on a competing entity and creates incentives for market fragmentation, since large competitors have little reason to rely on infrastructure operated by a rival. Multi-validator PoA distributes control, but exposes the infrastructure to governance stress, coordination failures and ambiguity over authoritative state, especially when operational governance of systems themselves become economically contested.

The analysis suggests that PoA is not unsuitable for financial DLTs per se, but is ill-suited where validation authority is exercised by competitors in the market rather than by a neutral public or institutionally separated operator.

Keywords: tokenization, proof-of-authority, consensus, blockchain, digital assets, smart contracts

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I. INTRODUCTION

Distributed ledger technology (DLT), including the tokenization of digital and real-world assets, is widely considered to be a major component of the future financial-market infrastructure, with potential applications in interbank settlement, cross-border payments, securities settlement, collateral management, and tokenized asset markets [1–4]. The major promise of DLT is the possibility of a shared programmable infrastructure capable of reducing reconciliation complexity between and within institutions, increasing automation through smart contracts, enabling atomic settlement of assets and payments, and potentially also operating continuously across jurisdictions, monetary zones and time zones. DLT-based systems are often expected to reduce settlement latency, lower operational costs, improve transparency and auditability, and simplify interoperability between fragmented financial infrastructures [3, 5]. These expectations have led to extensive experimentation by central banks, commercial banks, and infrastructure providers in areas such as wholesale central bank digital currencies (wCBDC), tokenized asset markets, securities settlement and cross-border payments [5, 6].

A number of central-bank-driven projects illustrate this development. Initiatives such as *Project Inthanon-LionRock*, its successor *Project mBridge* and *Project Cedar* investigate shared DLT infrastructures for cross-border payments and foreign-exchange settlement involving commercial and central banks [6–8]. More recently, *Project Agora* has aimed to improve cross-border correspondent banking by integrating tokenised commercial bank deposits and tokenised wholesale central bank money on a shared programmable platform while preserving the existing two-tier monetary system [9, 10]. Moreover, a growing number of industry-led projects aim to establish shared DLT infrastructures. Examples include *Fnality* [11], which explores tokenized settlement mechanisms backed by funds held at central banks, *Partior* [12], which focuses on cross-border interbank settlement and tokenized commercial-bank liabilities, and the *Canton Network* [13], intended as a "network of networks", aiming to provide synchronization infrastructure for tokenized financial-market applications. Also, the Commercial Bank Money Token proof of concept considers the European Public Network, a public-permissioned EVM-based infrastructure using PoA consensus for financial-sector tokenisation use cases [14]. XX Look into deloitte workshop sources for more real world projects XX

Although these projects differ substantially in architecture and governance, they illustrate a common pattern: transaction validation, ordering, or synchronization is typically restricted to predefined authorized entities rather than opened to anonymous public participation. This authority-based design choice is natural in regulated financial markets, but it creates a structural governance problem when authority over the ledger state is exercised by entities that are simultaneously competitors in the markets served by the infrastructure.

This discussion paper argues that Proof-of-Authority (PoA) and closely related authority-based consensus models are structurally problematic when used as the consensus layer of shared financial-market DLT infrastructures operated by commercial banks. The problem is not primarily technical, i.e., does not primarily concern common objectives for the design of consensus algorithms. The central issue is who is allowed to decide the ordering, inclusion, and finalization of economically significant transactions.

The analysis distinguishes two canonical configurations. In a single-validator setup, one commercial institution controls transaction ordering and validation, creating strategic dependence on a rival and incentives for market fragmentation. In a multi-validator setup, control is distributed across a consortium of commercial institutions, but the system becomes exposed to governance stress, coordination failures, and ambiguity about authoritative state under disputed conditions.

Related literature. This paper relates to two strands of literature. First, it contributes to the growing literature and policy discussion on DLT-based financial-market infrastructures, tokenisation, wholesale settlement, and programmable payment systems. Central-bank, policy, and regulatory work has examined the potential use of DLT in payment, clearing and settlement, the settlement of tokenised assets in central-bank money, cross-border wholesale CBDC platforms, and unified-ledger or programmable-platform architectures [2, 3, 5, 6, 8, 15–18]. Second, the paper builds on the technical work on distributed consensus, Byzantine fault tolerance and permissioned blockchains with particular focus on authority-based consensus mechanisms [19–26]. While this literature explains how distributed systems can agree on transaction validation, inclusion, ordering, and state replication, the present paper focuses on a different question, relating to relevant work on the future of financial-market infrastructure [1–3, 5, 27, 28]: whether the institutional allocation of authority to provide consensus is compatible with shared financial-market infrastructure when the validators are themselves competitors in the market. In this respect, the analysis also connects to work on transaction-ordering power, front-running and maximal extractable value, and to the financial-market-infrastructure literature on governance, legal basis, and settlement finality [29, 30].

Outline. The remainder of the paper is organized as follows. Section II defines PoA and related authority-based consensus models in the context of financial-market DLTs. Then, Section III analyzes single-validator and multi-validator configurations under commercial-bank operation. After a discussion of neutral governance and Real-Time Gross Settlement (RTGS) systems as an institutional benchmark in Section IV, we conclude in Section V.

II. AUTHORITY-BASED CONSENSUS IN FINANCIAL-MARKET DLTS

Conceptually, a DLT consists of a set of distributed nodes, which are computers or systems operated by participating entities, maintaining a shared ledger state through a consensus mechanism or algorithm [3, 31, 32]. The consensus algorithm describes how the nodes agree on the validity and ordering of transactions and ensures that participants converge on a common representation of the system state despite distributed operation. Consequently, the consensus mechanism effectively determines which transactions become part of the authoritative ledger state and under which conditions state transitions are finalized. Examples of distributed consensus mechanisms include classical distributed systems algorithms such as Paxos [20] or Raft [21] as well as blockchain-specific approaches such as Proof-of-Work (PoW)[32], Proof-of-Stake (PoS) [33], or Proof-of-Authority (PoA) (see, e.g., [24, 34, 35]).

Public DLT systems often rely on open participation and decentralized consensus at the cost of scalability, settlement latency and computational efficiency. In financial-market applications of DLTs, participants are supposed to be legally identified entities, primarily commercial banks and central banks, regulated financial-market infrastructures, and infrastructure providers. Naturally, the permissioned access to institutional DLT systems leads to the adoption of permissioned consensus mechanisms with predefined validators [15], in stark contrast to public networks like Bitcoin [32] or Ethereum [36], which originally popularized DLT.

Consequently, the consensus problem differs fundamentally from that of the public blockchains: rather than coordinating anonymous and potentially adversarial actors, the system must primarily ensure consistent state replication and transaction ordering among a fixed set of identifiable institutions that may cooperate operationally while remaining economically and strategically distinct. This distinction is central for financial-market DLTs. The relevant trust problem is not eliminated by legal identification of validators, but transformed into a question of how ordering and validation authority is allocated among institutions with potentially divergent incentives.

Proof-of-Authority and centralized consensus. In this paper, PoA is used in a broad sense to denote permissioned consensus mechanisms in which transaction ordering, validation, or finality depends on a legally identified and administratively selected set of authorities. Some variants of PoA also allow validators to vote about the introduction of new validators according to predetermined rules. This includes explicit Ethereum-derived PoA protocols such as Clique, which is an Ethereum PoA protocol specified in EIP-225 [24], and Byzantine-fault-tolerant authority-based protocols such as IBFT and IBFT 2.0 [25, 26], but also functionally equivalent authority-based arrangements in commercial-bank DLT systems, such as notary services (see, e.g., [37]), ordering services (as in [38]), validator committees, or synchronization or sequencing committees [13]. Enterprise Ethereum clients such as Hyperledger Besu and GoQuorum implement PoA-style protocols including Clique, IBFT 2.0, and QBFT for private permissioned networks [39, 40].

The purpose of this terminology is not to claim that all such systems implement the same protocol. Rather, PoA is used as an analytical proxy for restricted consensus authority in systems where a selected set of legally identifiable institutions determines the authoritative ledger state.

The appeal of the PoA approach is clear. By eliminating open participation and competition from the consensus, PoA enables high throughput, low latency, and operational simplicity due to the reduction of the coordination problem, some of the most prominent drawbacks of early public DLT networks, while still providing advantages such as programmability and atomic settlement. These technical advantages do not resolve the institutional allocation problem. In financial-market infrastructures, the relevant question is not only whether a restricted validator set can process transactions efficiently, but whether the institutions controlling ordering and finality have incentives compatible with neutral market infrastructure.

Ultimately, the role of validators is economically significant, since they determine the ordering and finalization of payments, securities transfers, and other high-value financial transactions. Transaction ordering and inclusion are particularly relevant because it determines the underlying system state under which transactions are executed. This directly affects the economic outcome of transactions and may result in certain operations to be delayed, reordered, or even rejected depending on the effects of the evolving ledger state. The literature on front-running and maximal extractable value in the crypto space illustrates that control over ordering can itself become economically valuable, see, e.g., [29, 30].

Thus, the governance structure underlying authority-based consensus in financial-market DLT systems is the central institutional design problem. Permissioned validation identifies who may participate in consensus, but it does not by itself determine how validation authority should be allocated among participating institutions in the long term. This question is not merely technical. It reflects the economic structure of the underlying market and becomes more contested as the value of assets processed on the platform grows.

For the following analysis, two canonical configurations are distinguished: single-validator PoA and multi-validator PoA. These configurations do not exhaust all possible institutional designs, but they capture the main authority structures relevant for commercial-bank-operated PoA and related permissioned consensus systems. It determines whether the system is governed by unilateral control or by collective agreement among competitors, and thus directly affects participation incentives, robustness of consensus, and the (legal) interpretation of ledger state. The next section analyzes the structural failure modes associated with both configurations.

III. STRUCTURAL FAILURE MODES OF SHARED MARKET INFRASTRUCTURE UNDER COMMERCIAL-BANK VALIDATION

The following analysis concerns DLTs with authority-based consensus layer operated by commercial banks and intended to function as shared financial-market infrastructures among multiple commercial banks. Rather than focusing on specific implementations, the analysis abstracts from existing projects to their underlying consensus structures and closely examines the effects of centralization in the economically relevant transaction ordering in commercial-bank DLT systems. The problem with PoA in commercial-bank-operated systems is not that the mechanism makes such systems technically unusable. Rather, the problem is that its trust model does not match the institutional structure of the competition in financial markets. A PoA ledger must answer a simple question: who is allowed to decide the canonical ordering of economically relevant transactions?

Figure 1 summarizes the institutional incentive problems that arise from different allocations of validation

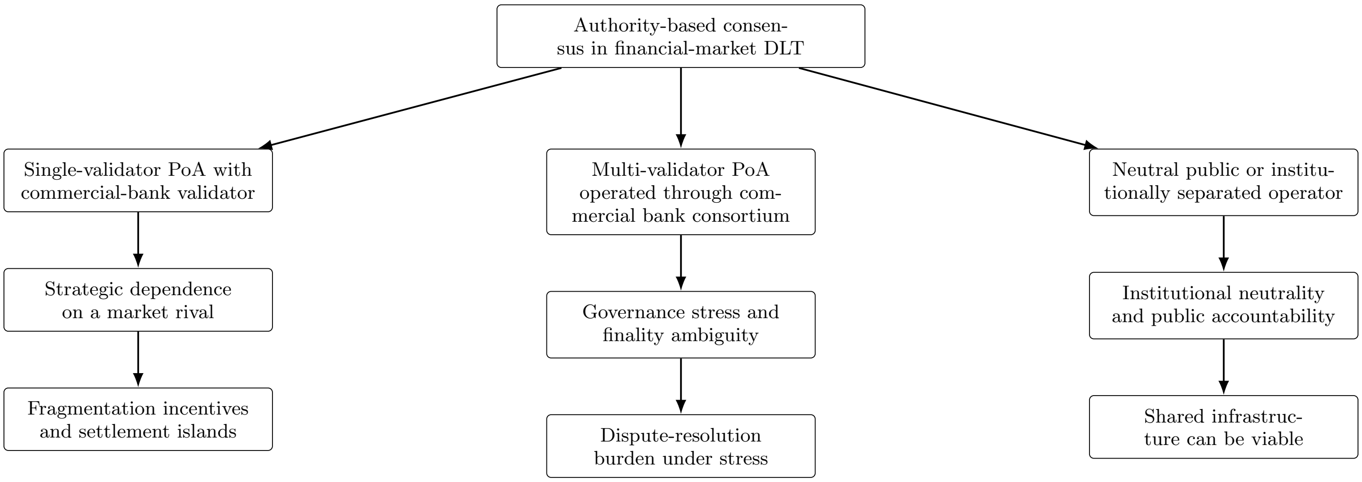

FIG. 1. Institutional failure modes of authority-based consensus in shared financial-market DLT infrastructures. The figure illustrates that the relevant distinction is not only technical decentralization, but the institutional position of the authority controlling ordering, inclusion, and finality.

authority in commercial-bank-operated DLT infrastructures. The figure highlights the basic logic. A single commercial-bank validator creates asymmetric dependence on a rival. A commercial-bank validator consortium distributes control, but shifts the problem into collective governance among competitors. Neutral governance changes the institutional position of the authority, but requires a public mandate, legal basis, or institutional separation from commercial competition. In both cases, the supposed simplicity of PoA reintroduces coordination and trust problems in institutions that consensus in DLTs shall originally resolve. We analyze the two different cases in depth in the following.

A. Single-validator Proof-of-Authority: centralization and market fragmentation

In the single-validator case, PoA collapses into a centrally operated ledger. This may be acceptable when the validator is a central bank, public authority, or legally mandated infrastructure operator without being a competitor in the financial markets. It is much less plausible when the validator is a commercial bank. A major bank is unlikely to rely on a ledger whose transaction ordering, inclusion, and operational availability are controlled by a direct competitor.

The problem is not solely reputational. A single validator has privileged technical power over the ledger state. It can delay transactions, censor transactions, reorder transactions, or observe commercially sensitive flows before they are finalized. As discussed above, transaction ordering by itself can significantly affect the ledger state under which transactions are executed and can be exploited for the purpose of the validator, see, e.g., [29, 30]. Even if selfish reordering and further malicious behavior is legally prohibited, the technical possibility creates a strategic dependency on the validating institution. The concern is particularly relevant in high-value financial-market infrastructures, where transaction ordering, inclusion, and timing can have direct economic consequences.

Legal accountability alone does not remove this problem. Centrally operated blockchain systems can in principle achieve high performance and process large transaction volumes [41, 42]. In such settings, large amounts of economic value may depend on transaction ordering and inclusion decisions. The volume and frequency of transactions would make it difficult for any jurisdiction to monitor, detect, and adjudicate manipulative behavior fully ex post. Legal enforcement is therefore an important safeguard, but cannot substitute the neutral governance of the consensus layer itself.

Consequently, sufficiently large competitors would have incentives either not to participate or to establish rival infrastructures. The likely result is market fragmentation: instead of converging on shared infrastructure, the market may split into bank-operated settlement islands. Interoperability would then have to be restored through bridges, APIs, contractual arrangements, or reconciliation mechanisms, reintroducing the operational complexity and liquidity fragmentation, i.e., precisely the frictions that shared DLT infrastructure promise to reduce. In such a scenario, the DLT-specific value proposition becomes weak at the inter-institutional level. The systems begin to resemble an updated form of internal replicated databases, with benefits concentrated mainly in interactions between dominant platform operators, the large market participants, and smaller participants.

B. Multi-validator Proof-of-Authority: consortium governance and finality ambiguity

The multi-validator case appears more attractive at first sight, since validation authority is distributed across several institutions rather than concentrated in a single entity. In such a setup, validation is typically either exercised by a consortium of authorized institutions or rotates among them according to predefined protocol rules, such as, e.g., round-robin alternation of block production. These setups reduce unilateral control, but they do not remove the institutional problem. Rather, the dependence on one commercial actor is replaced by collective governance among competing institutions.

Permissioned multi-validator systems may rely on Byzantine-fault-tolerant protocols, quorum rules, proposer rotation, or tightly governed validator committees to prevent persistent forks, that are well-known from the crypto space. These mechanisms can be effective under normal operating conditions, but they do not determine validator admission, software upgrades, sanctions, emergency interventions, dispute resolution, or the relationship between technical and legal finality. This is not handled by the consensus protocol itself, but outsourced to the governance framework of the consortium. The absence of routine forks is therefore not sufficient: the critical question is who controls the procedures that determine the authoritative state when the protocol, the participants, or the legal interpretation of ledger events come under stress. The governance choices may affect access, liquidity, transaction timing, and the distribution of rents within the infrastructure, and are therefore not merely technical parameters, but economically relevant institutional decisions. All rules determined through the governing framework may become economically contested, and the infrastructure may be exposed to governance stress, coordination failures, or, in extreme cases, competing claims about the authoritative ledger state.

The governance burden further increases with the number and heterogeneity of participants. All changes to the DLT system's governance and operation including technical upgrades, validator rules, operational procedures, and emergency responses, must be coordinated among institutions with partly divergent commercial interests. In extreme cases, economically contested conflicts may even lead to governance deadlock. Legal frameworks for these systems must therefore specify not only normal operation, but also failure handling, validator disputes, emergency coordination, and the relationship between technical finality and legal finality. This last issue is not unique to PoA: the relationship between technical finality and legal finality must be addressed for any programmable DLT system representing legally recognized assets. Where such systems process high-value payments, securities, or tokenized claims, unresolved finality disputes may create spillovers beyond the platform itself and, in sufficiently large or interconnected infrastructures, may even become a source of systemic risk.

A separately incorporated consortium or jointly governed operating entity may mitigate some frictions by acting as the formal operator of the consensus layer. Yet the consortium remains shaped by the interests, voting rights, and strategic influence of its members. New entrants may also face rules and technical choices established before they joined and which they cannot easily influence ex post. Open and neutral market participation is therefore not guaranteed. Thus, market fragmentation may still occur at a larger scale, but similar to the single-validator case, if the consortium's governance arrangements are perceived as biased or exclusionary by some market participants.

The relevant failure mode of multi-validator PoA is, consequently, not simple centralization, but institutional ambiguity. Control is distributed, but the authority to resolve disagreement, define finality, and adapt the system remains embedded in a consortium of competing commercial actors. For high-value financial-market infrastructures, this creates governance and legal-finality risks that cannot be solved by the consensus protocol alone.

IV. NEUTRAL GOVERNANCE AS THE INSTITUTIONAL CONDITION FOR AUTHORITY-BASED CONSENSUS

How neutrality changes the institutional problem of PoA. The failure modes identified above do not arise from authority-based consensus as such, but from the institutional position of the authority exercising control over ordering, inclusion, and finality. In a commercial-bank-operated PoA system, authority is assigned to actors that are themselves competitors in the market served by the infrastructure. Neutral governance changes the problem because the authority is separated from the competitive interests of the participants using the infrastructure. This does not eliminate the need for accountability, operational resilience, or clear governance procedures, but it removes the central source of strategic mistrust that makes commercial-bank-operated PoA unsuitable for shared financial-market infrastructures. If the validating entity is a public authority, an independent operator, or another institutionally neutral body, the dependence on authority no longer takes the form of dependence on a market rival. Under neutral governance, authority-based consensus can therefore function as a technical mechanism providing non-discriminatory ordering, inclusion, and finality within the infrastructure. In such a setting, the potential benefits of DLT, including atomic settlement and programmability through smart contracts, can flourish.

In practice, the neutrality may be provided by central banks, whose public mandate and established role in settlement finality make them institutionally well suited to operate or govern authority-based financial-market infrastructures, but whose direct involvement may limit flexibility, innovation speed, or private-sector experimentation. Alternatively, neutrality may be provided by independent technical service providers, which can offer operational expertise and greater implementation flexibility, but only if their role is confined to technical operation and embedded in governance arrangements that prevent capture by, or preferential treatment of, individual market participants. A technical outsourcing arrangement alone does not create neutrality if ultimate control remains with a subset of competing market participants.

RTGS systems: Authority-based settlement infrastructures with neutral governance RTGS systems illustrate that an authority-based settlement infrastructure can be compatible with financial-market infrastructure where the operator is neutral, has a public mandate, provides or settles in risk-free central bank money, and is supported by clear legal finality and public accountability. RTGS systems, therefore, serve as an institutional benchmark rather than a technical analogy to DLT consensus. They show that centralization of the authoritative settlement layer is not necessarily problematic in itself. The decisive condition is that the authority is not a commercial competitor in the markets served by the infrastructure.

The fact that existing RTGS systems are not DLT systems does not weaken this comparison, because the argument concerns the institutional allocation of settlement authority rather than the specific technical architecture used to record transactions. The relevant lesson is therefore not that DLT-based systems should reproduce the database architecture of RTGS systems. Rather, the lesson is that the authoritative settlement layer of a high-value financial-market infrastructure requires neutral and legally reliable governance.

The institutional suitability of RTGS systems can be summarized by the following features, each of which addresses one of the strategic vulnerabilities that arise when authority is assigned to competing market participants:

- **Neutral settlement authority:** The central bank is not a commercial competitor; it has no incentive to exploit transaction ordering, access decisions or information asymmetries.
- **Legal finality:** Settlement takes place in central bank money and is supported by clear legal frameworks; this removes ambiguity about the authoritative settlement state.
- **System-wide infrastructure:** RTGS systems provide a common settlement layer for high-value payments and related financial-market activity. Their role as shared infrastructure reduces incentives for fragmentation into rival settlement systems.
- **Trusted governance:** Central bank mandate (monetary and financial stability, payment system efficiency) aligns with system-wide interests rather than private incentives.
- **Operational responsibility and accountability:** Authority over the infrastructure is matched by identifiable responsibility for system operation, risk management, incident handling, and crisis procedures.

The limited role of privately operated RTGS systems. The limited role of private operation in RTGS systems supports this point. Existing arrangements do not generally assign autonomous settlement authority to a commercial bank or a subset of competing commercial banks. Where private entities are involved, their role is typically embedded in central-bank settlement, public oversight, or delegated technical operation. The Swiss SIC system illustrates this distinction: it is operated by SIX Interbank Clearing Ltd on behalf of the Swiss National Bank (SNB), while the SNB remains the relevant public authority behind the system [43, 44]. Similarly, the historical CHAPS arrangement in the United Kingdom involved private-sector governance of the high-value payment system, but this model was replaced in 2017 by direct operation of CHAPS by the Bank of England in order to strengthen end-to-end risk management and financial stability [45, 46].

Other high-value payment systems reinforce the same distinction. T2 and Fedwire Funds are central-bank-operated RTGS systems [47, 48]. Lynx is operated by Payments Canada but relies on settlement accounts at the Bank of Canada [49]. CHIPS and EURO1, although privately operated high-value payment systems, are not autonomous private RTGS systems; they rely on liquidity-saving netting or hybrid settlement mechanisms and remain anchored in central-bank money or central-bank-operated settlement infrastructures [50, 51]. Thus, existing real-world systems support the central claim of this section: authority-based consensus is not unsuitable for financial-market infrastructures per se; it is unsuitable when the authority is itself a competing market participant or coalition of competing market participants.

V. CONCLUSIONS

The contribution of this paper is to shift the assessment of permissioned DLT consensus from technical performance alone to market structure, governance, and legal finality. In commercial-bank-operated financial-market infrastructures, the central design problem is not merely how consensus is reached, but who is institutionally authorized to determine the canonical ordering and final settlement state. Proof-of-Authority (PoA) consensus is therefore not only a technical design choice. It allocates control over transaction inclusion, ordering, and finality to an identifiable authority or group of authorities whose institutional position matters for the viability of the infrastructure.

This paper has argued that PoA consensus is structurally ill-suited for DLT-based financial-market infrastructures operated by market participants despite their technical performance advantages. For commercial-bank-operated PoA systems, structural institutional problems are identified. A single-validator design concentrates settlement authority in one commercial bank and thereby creates strategic dependence on a market rival. A multi-validator design distributes this authority, but does not make it neutral; instead, it shifts the problem into consortium governance, where competing validators may face disputes over protocol rules, validator conduct, software upgrades, emergency interventions, and legal finality. Consequently, the weakness of commercial-bank-operated PoA is therefore not that it cannot establish technical consensus, but that it does not provide an institutionally robust basis for a fair, open, and legally reliable shared financial-market infrastructure.

Discussion. The analysis does not imply that PoA consensus is unsuitable for all financial DLT applications. Rather, it identifies the institutional conditions under which authority-based consensus can be viable. Authority-based consensus may be appropriate where the validating authority is not itself a competing market participant, where governance is embedded in a public mandate, or where the ledger does not serve as a common settlement infrastructure for rival institutions. This distinction explains why central-bank-operated, central-bank-led or otherwise neutrally governed DLT arrangements are more plausible than commercial-bank-operated systems for high-value settlement.

The broader implication is that consensus mechanisms for financial-market infrastructures cannot be assessed independently of institutional design. In high-value settlement systems, the consensus layer allocates authority over inclusion, ordering, and finality, and therefore determines who can produce the authoritative state of the system. DLT-based financial-market infrastructures therefore require a careful separation between technical validation and institutional authority. Efficient consensus is not sufficient; the infrastructure must also provide a governance structure capable of producing a neutral, legally reliable, and final settlement state, potentially embedded within the consensus mechanism itself.

As a consequence of this conclusion, hybrid DLT architectures separating public settlement authority from private innovation using a multi-layer approach may be an interesting design direction to explore for application in financial-market infrastructures. One could imagine a design where a layer-one (L1) infrastructure operated or governed by a public authority could provide the neutral settlement, synchronization, and finality layer, while privately operated layer-two (L2) networks could offer application-specific functionality, market services, or operational innovation. This would allow interbank settlement of high-value obligations to remain anchored in a neutral institutional framework, while still enabling market participants to develop DLT-based services on top of that foundation on their respective L2s. In a setting where central banks provide such an L1, one could explore the possibility of issuing and settling in tokenized wholesale central bank money via such an infrastructure.

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Appendix A: Potential figures for main paper

Configuration	Authority structure	Immediate benefit	Institutional concern
Single commercial-bank validator	One competing market participant controls transaction ordering and validation	Operational simplicity, low coordination overhead, high performance	Asymmetric dependence on a rival, potential censorship or reordering power, platform dependence for smaller participants, and fragmentation incentives for sufficiently large competitors
Commercial-bank validator consortium	Multiple competing institutions jointly determine the canonical ledger state	Distribution of authority across participants, reduced unilateral control	Coordination failures, governance disputes, fork or finality ambiguity, complex ex-post legal resolution
Neutral governance through public validator or independent technical service provider	A non-commercial authority or independent service provider operates or governs the authoritative settlement layer	Clear accountability, unified participation, legally supported finality	Requires public mandate, legal basis, and institutional neutrality; not replicable by ordinary commercial-bank governance

TABLE I. Canonical authority-based consensus configurations in financial-market DLT infrastructures. The relevant distinction is not only the number of validators, but whether validation authority is exercised by a neutral public operator or by commercially competing market participants. The classification is specific to PoA and related authority-based consensus models and is not intended as a general taxonomy of all consensus mechanisms. The table does not imply that central-bank validation is necessarily appropriate for every conceivable shared financial-market DLT infrastructure.

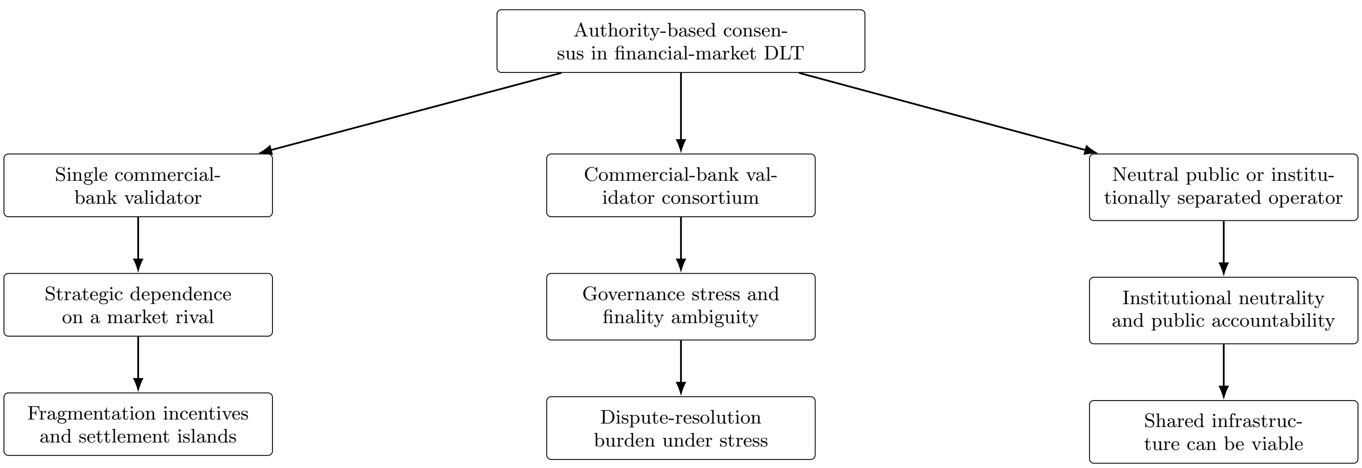

FIG. 2. Institutional failure modes of authority-based consensus in shared financial-market DLT infrastructures. The figure illustrates that the relevant distinction is not only technical decentralization, but the institutional position of the authority controlling ordering, inclusion, and finality.

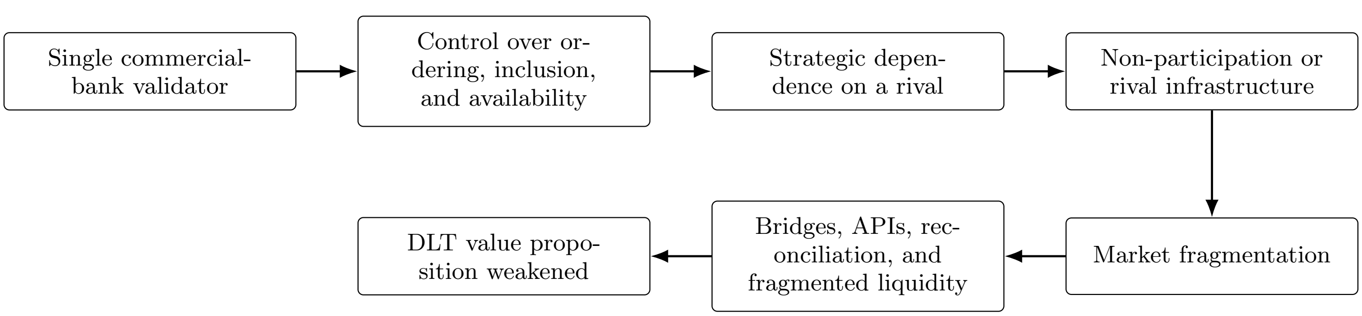

FIG. 3. Causal mechanism from single-validator commercial-bank PoA to market fragmentation. A commercial bank acting as validator creates strategic dependence for rival institutions, which may induce non-participation or the development of rival infrastructures.

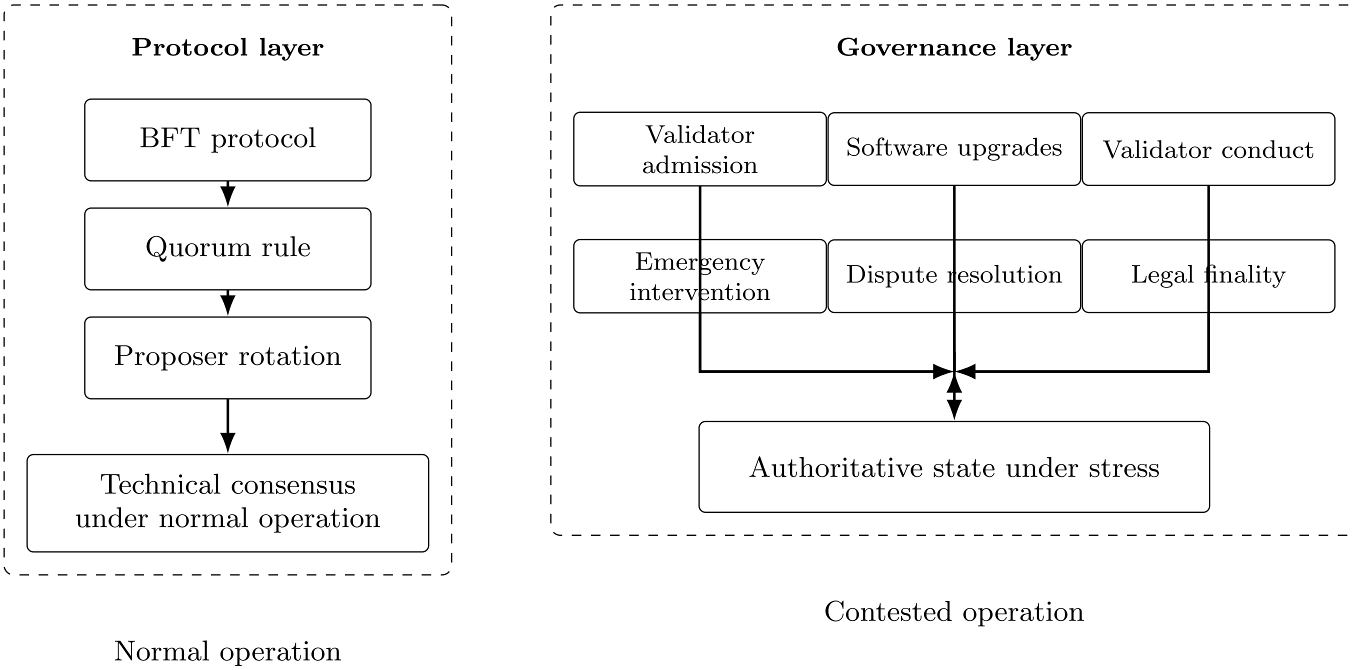

FIG. 4. Technical consensus and institutional finality in multi-validator PoA. Consensus protocols may determine state transitions under normal operation, but contested cases require governance procedures for admission, upgrades, validator conduct, emergency intervention, dispute resolution, and the relation between technical and legal finality.

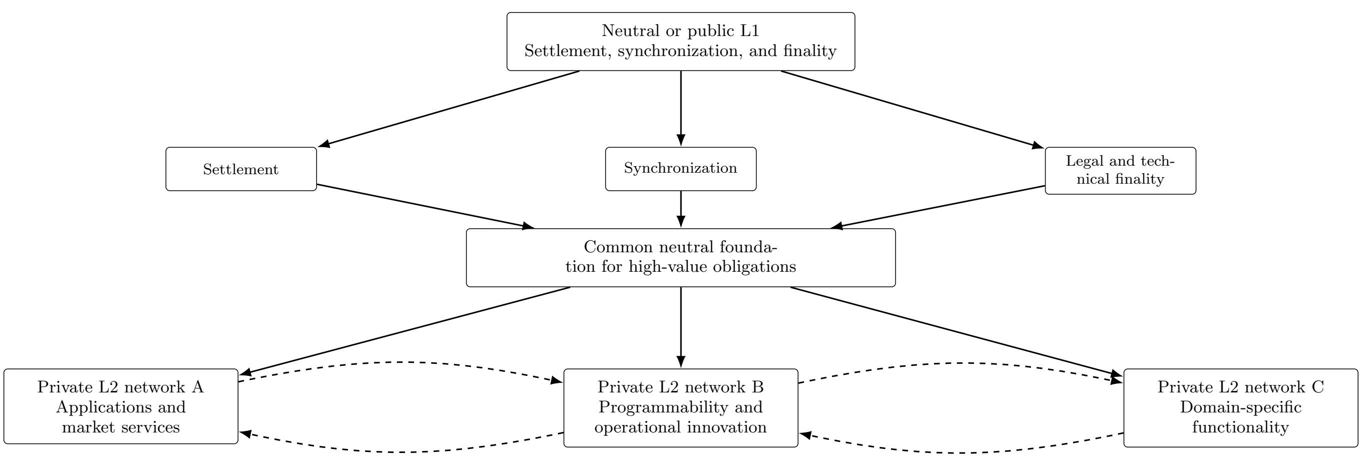

FIG. 5. Hybrid architecture separating neutral settlement authority from private DLT-based innovation. A neutral L1 could provide the authoritative settlement, synchronization, and finality layer, while private L2 networks provide application-specific services on top of that foundation.